

FCC 3101

CHAPTER 1

**ORIENTATION TO
CIVIL ENGINEERING**

1.1 ORIGINS OF ENGINEERING

1.1.1 As **civilization** took place, the human race realized that it was possible to overcome natural challenges and thereby improve on their living standard.

1.1.2 Some of the initial challenges were: -

1. **Shelter** from bad weather and wild animals
2. **Transport** across valleys and rivers
3. **Tools** to make their daily work easier
4. **Arms** to defend themselves

1.1.3 According to history **recorded by the Englishman** engineering originated in the Middle East.

1.1.4 Examples of early engineering works include: -

1. The **Pyramids** of Egypt constructed during the **bronze age** (5000 to 1000 BC).
2. The **wedge, lever, wheel, pulley, inclined plane** and **screw (simple machines)** invented during the **bronze age** (5000 to 1000 BC).
3. The **Roman Aqueducts** constructed during the **middle ages** (50 BC to 1400 AD).
4. The **cable stayed bridge** constructed towards the end of **renaissance** (1600 AD).



Figure 1.1 Pyramids of Egypt



Figure 1.2 Roman Aqueducts

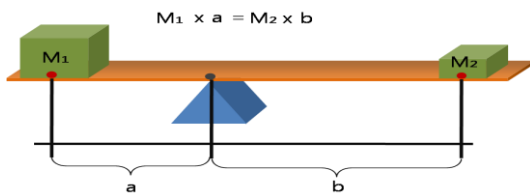


Figure 1.3 The Lever

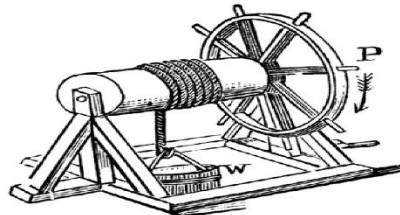


Figure 1.4 The Wheel

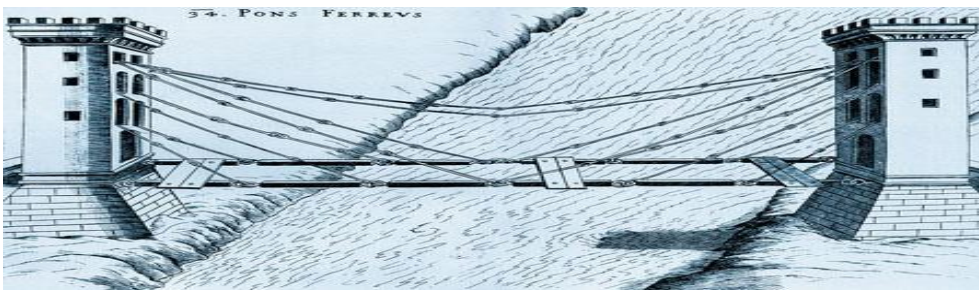


Figure 1.6 Cable Stayed Bridge

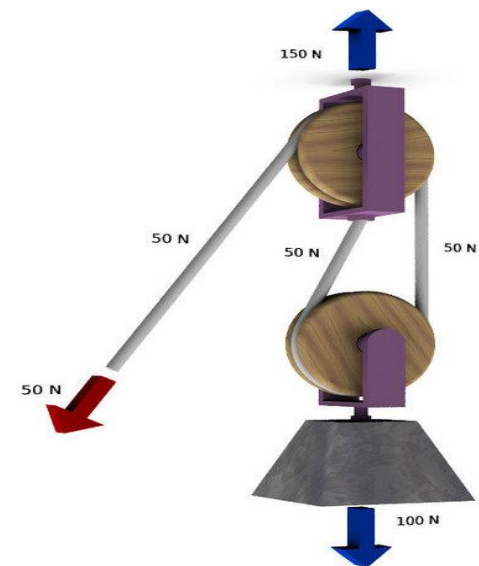


Figure 1.5 The Pulley

1.2 ADVANCES IN ENGINEERING

Over the centuries, engineering has advanced to a level where: -

- 1.2.1 Every single development is attributed to engineering.
- 1.2.2 Every tool, equipment and machine that we use is a product of engineering.
- 1.2.3 Every aspect of life is made better by engineering.
- 1.2.4 Without engineering, life would be primitive.



Figure 1.7 Interchange in Shanghai



Figure 1.8 Airbus A380



Figure 1.9 The Burj Khalifa

1.3 STEM SUBJECTS

Engineering is one of four cluster subjects referred to as **STEM**: -

- 1.3.1 Science** is the motherboard on which all engineering knowledge is based.
- 1.3.2 Technology** is the collection of techniques used to solve day-to-day problems.
- 1.3.3 Engineering** is the highest level of technology used to solve major challenges.
- 1.3.4 Mathematics** is the common tool which is necessary in the other subjects.

1.4 BRANCHES OF ENGINEERING

- 1.4.1 Due to the many challenges facing the human race, and because of advances in science and technology, the **engineering profession** has evolved into many several **disciplines** referred to as **branches of engineering**. Each branch deals with a particular set of engineering aspects, but the branches are closely inter-related.
- 1.4.2 Some of the branches of engineering are discussed below. However, the focus of **FCC 3101** is to introduce the “fresher” to **Civil Engineering**. [The other branches of engineering will remain suppressed for the remainder of this chapter and for the rest of **FCC 3101**].

1.4.3 The **classical** and **most important** branches of engineering are: -

1. **Civil Engineering:** Civil Engineering deals with buildings and structures, roads, highways and railways, airports and seaports, water supply systems, waste water treatment, flood mitigation and control, irrigation and canals, tunneling and underground works.
2. **Mechanical Engineering:** Mechanical Engineering deals with tools, equipment, machines, plant, industry, manufacturing and processing.
3. **Electrical Engineering:** Electrical Engineering deals with electric power generation, power supply and distribution, electronic equipment, electronic machines, telecommunications and information systems.

1.4.4 Emerging branches of engineering include: -

1. **Chemical Engineering:** Chemical Engineering deals with the design and operation of chemical processing plants which are used to convert raw materials into semi-finished products or finished products.
2. **Mechatronic Engineering:** Mechatronic engineering is an interdisciplinary branch of engineering which integrates mechanical engineering, electronics engineering and computer science to design robotic equipment which use artificial intelligence to operate themselves.
3. **Biomedical Engineering:** Biomedical Engineering is a branch of engineering which applies engineering principle to design and operate biomedical equipment necessary for specialized healthcare.

1.5 CIVIL ENGINEERING

- 1.5.1 **Civil Engineering** is a **discipline**, within the **engineering profession**, that deals with the design, construction, maintenance and safe demolition of infrastructural facilities such as: -
1. **Buildings:** Domestic houses, commercial buildings, industrial sheds, institutional buildings, sports stadia and gymnasias, bunkers and silos.
 2. **Transport:** Roads, highways, railways, seaports and airports.
 3. **Bridges:** Foot bridges, highway bridges, interchanges, overpasses and underpasses.

4. **Geotechnical:** Tunnels, dams, hydroelectric power stations and underground works.
5. **Water Resources:** Water intake, treatment, storage, supply and distribution systems.
6. **Public Health:** Waste water and solid waste collection, treatment and recycling.
7. **Public works:** Flood mitigation and control, storm drainage and irrigation canals.

1.5.2 Civil Engineering is considered to be the oldest engineering discipline.

1.5.3 The term Civil Engineering was coined sometimes in the 18th century (1700 – 1799) so as to distinguish “**civilian**” engineering from “**military**” engineering.

- 1.5.4 The first university level institution to teach Civil Engineering was *École Nationale des Ponts et Chaussées* (**National School for Bridges and Roadways**) which was founded in France in 1747.
- 1.5.5 The first **self-proclaimed** civil engineer was **John Smeaton** (1724 – 1792), a Briton, who was **responsible** for the design and construction of bridges, canals, harbors and lighthouses in many countries in Europe.
- 1.5.6 **John Smeaton** is also credited with efforts to form the first society of civil engineers.

- 1.5.7 By the beginning of the 19th century (1800 – 1899), Civil Engineering had been established in many other European countries, including the United Kingdom and the United States.
- 1.5.8 Civil Engineering is based on the **application of scientific principles and mathematical tools** to analyse **(diagnosis)**, design **(prescription)** and construct **(therapy)** the desired facility.
- 1.5.9 Civil Engineering is therefore defined as the “**art and science of harnessing natural resources for the use, comfort and convenience of mankind**”.



**Are You Properly Oriented
In Civil Engineering?**